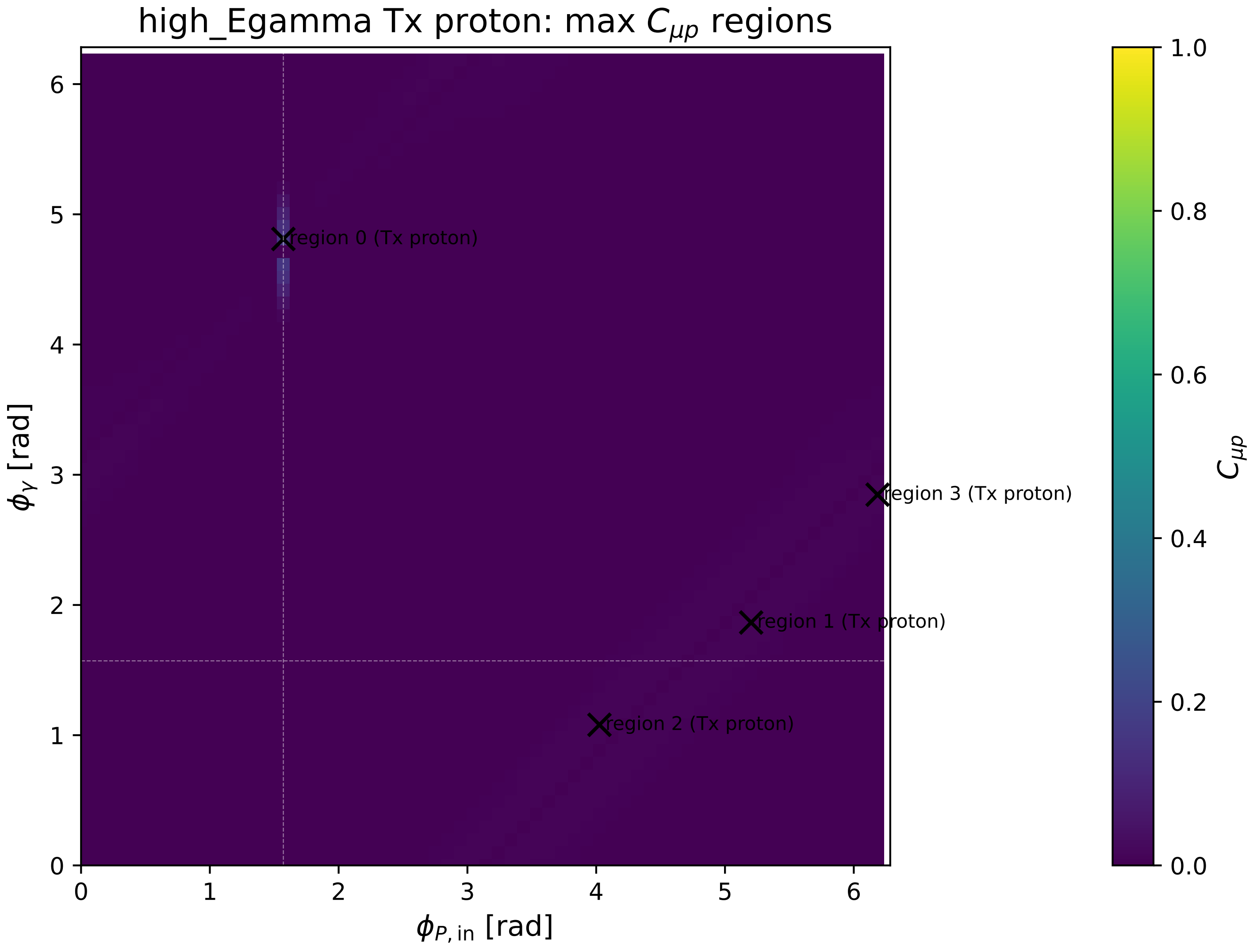
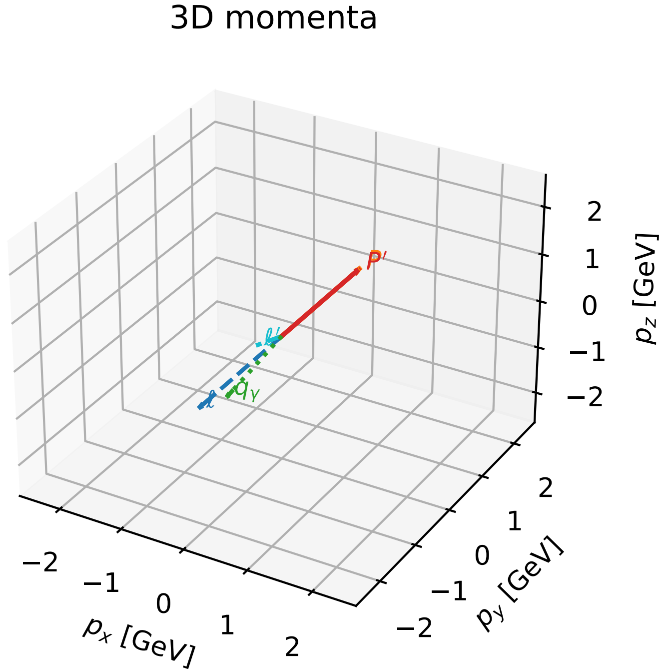

high_Egamma Tx proton: max $C_{\mu p}$ regions
region 0 (Tx proton)
region 1 (Tx proton)
region 2 (Tx proton)
region 3 (Tx proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\mu p}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Tx proton characteristic max $C_{\mu p}$ configuration

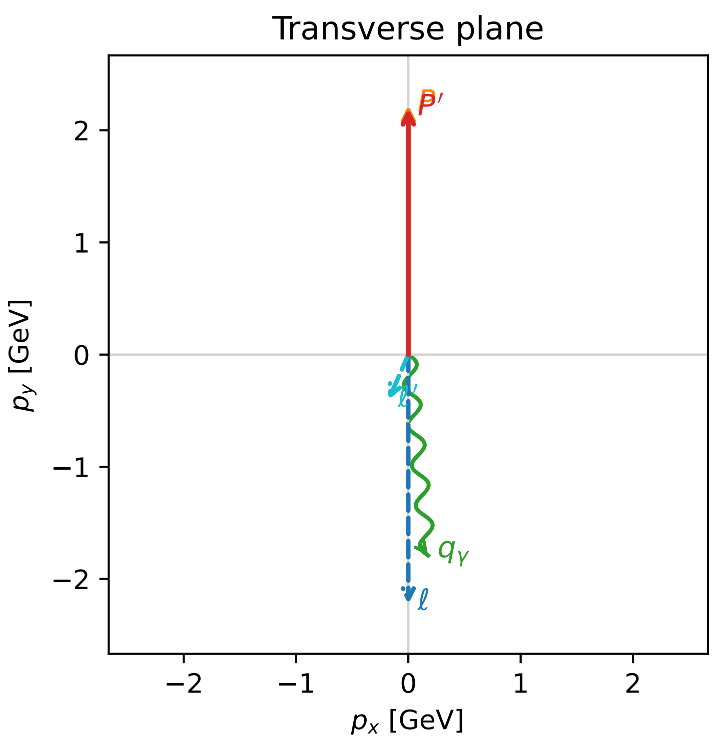


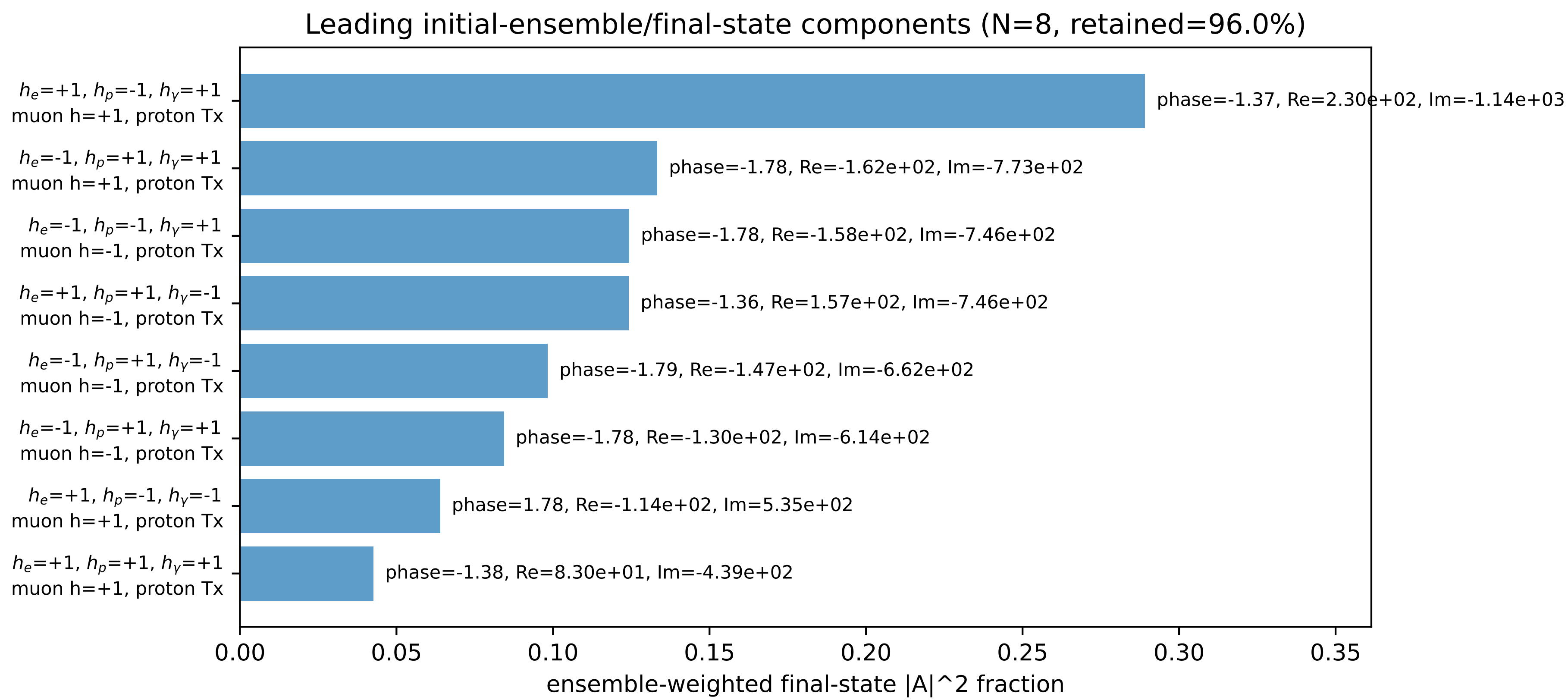

```
c_ep_high_egamma_tx_proton_region_0  C_{mup}=0.152555
region: high_Egamma_Tx_proton_region_0  final-pair delta_xy=2.72877 rad
outgoing-state purity: 0.512094
kinematic point: high_s_high_theta_in_high_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_\mu}\rho(h_\mu, T_{x,p})$

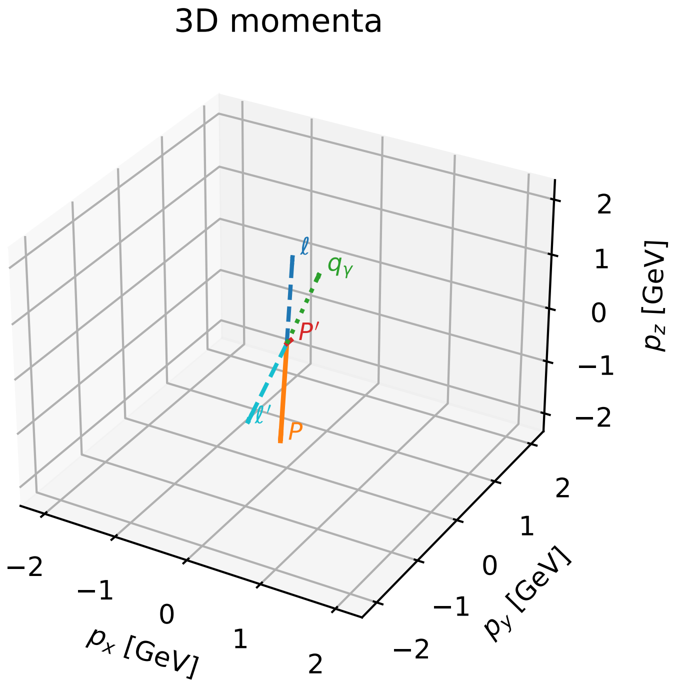
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.19415
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=1.8, ϕ_γ=4.81056
Q^2=0.199845, x_B=0.0120018, t=-0.000128163, W^2=17.3312, y=0.807338
$\theta(\ell', \gamma)$=29.278 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -0.0000 p_y= -2.2231 p_z= -0.0000
P E= 2.4129 p_x= 0.0000 p_y= 2.2231 p_z= 0.0000
ℓ' E= 0.4523 p_x= -0.1764 p_y= -0.4028 p_z= 0.0000
P' E= 2.3862 p_x= 0.0000 p_y= 2.1942 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.1764 p_y= -1.7913 p_z= 0.0000





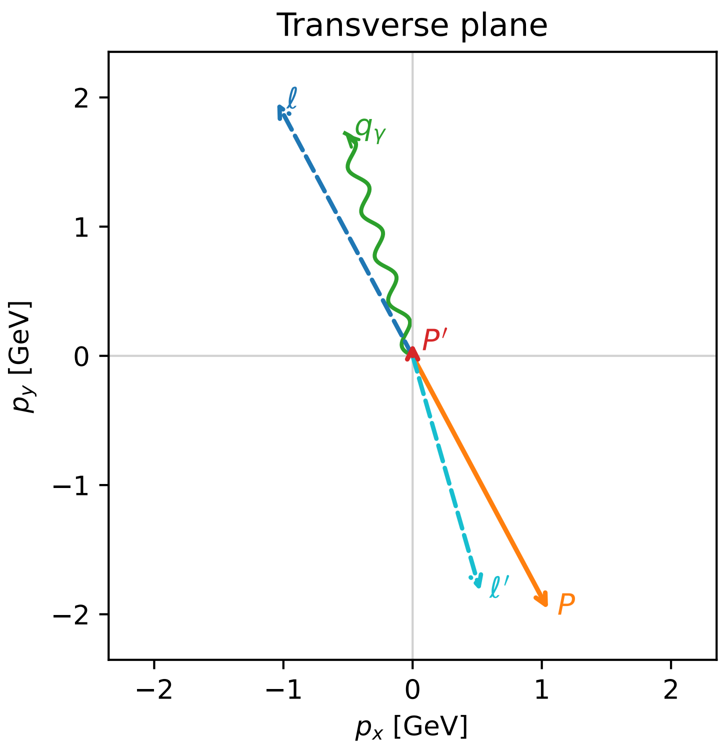
Tx proton characteristic max $C_{\mu p}$ configuration

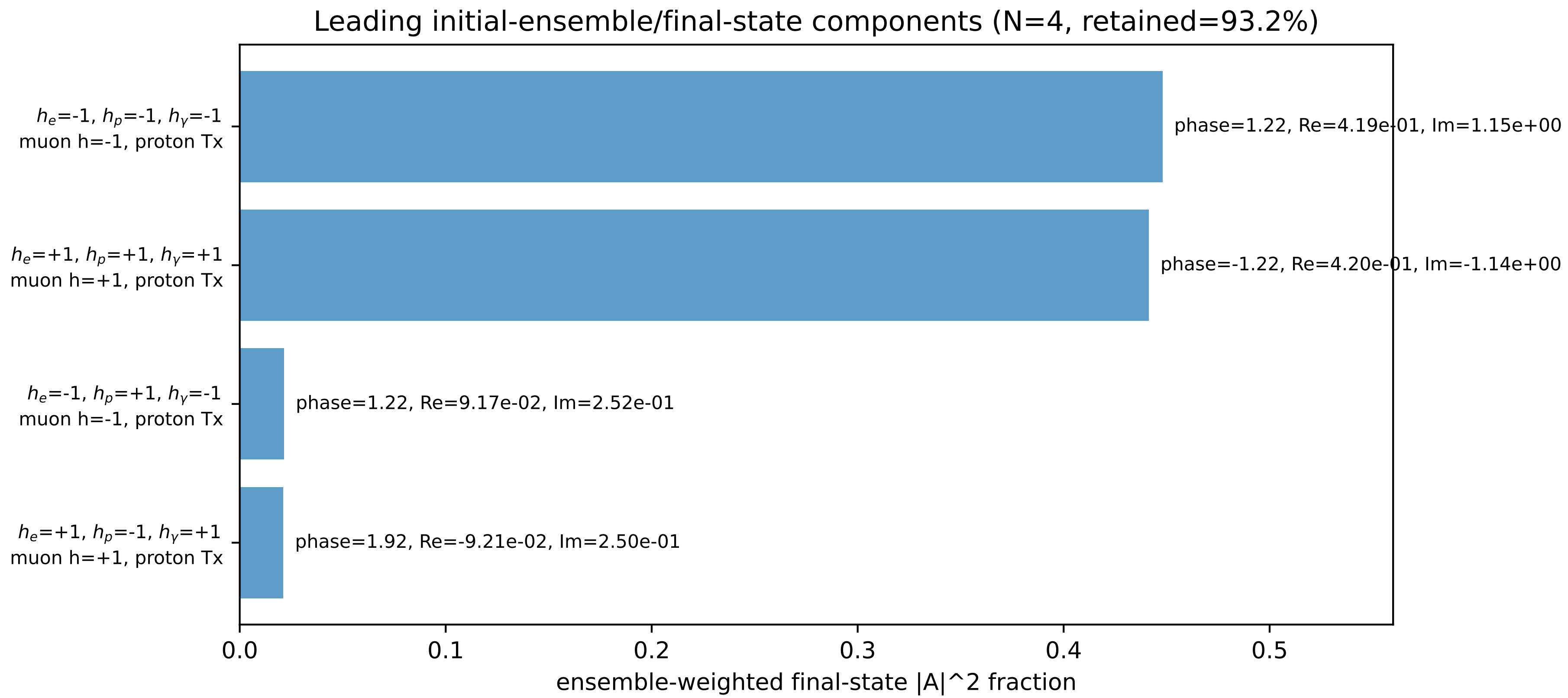


c_ep_high_egamma_tx_proton_region_1 C_{mup}=0.00942741
region: high_Egamma_Tx_proton_region_1 final-pair delta_xy=2.86188 rad
outgoing-state purity: 0.50002
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_\mu}\rho(h_\mu, T_{x,p})$

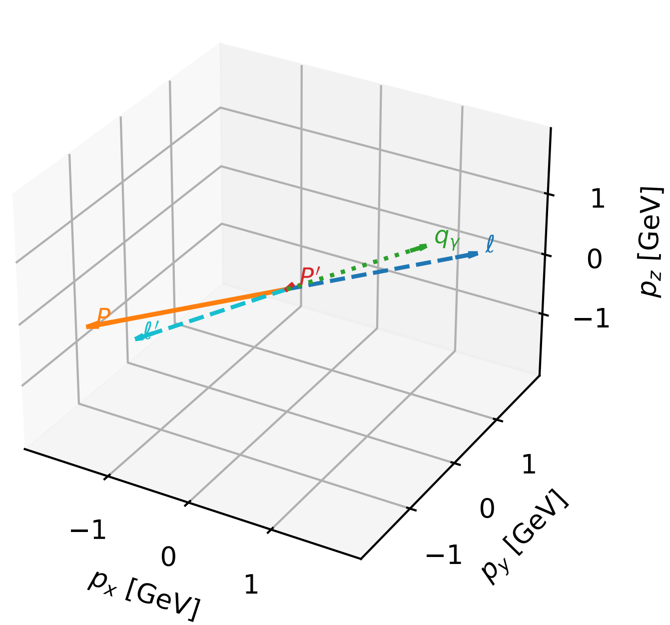
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=0.0965635
θ_{in}=1.57079, ϕ_ℓ=2.06167, ϕ_P=5.20326
E_γ=1.8, ϕ_γ=1.86532
Q^2=16.6433, x_B=0.844611, t=-3.16948, W^2=3.94183, y=0.955417
$\theta(\ell', \gamma)$=179.151 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -1.0480 p_y= 1.9606 p_z= -0.0000
P E= 2.4129 p_x= 1.0480 p_y= -1.9606 p_z= 0.0000
ℓ' E= 1.8956 p_x= 0.5225 p_y= -1.8191 p_z= 0.0000
P' E= 0.9430 p_x= 0.0000 p_y= 0.0966 p_z= 0.0000
q_γ E= 1.8000 p_x= -0.5225 p_y= 1.7225 p_z= 0.0000





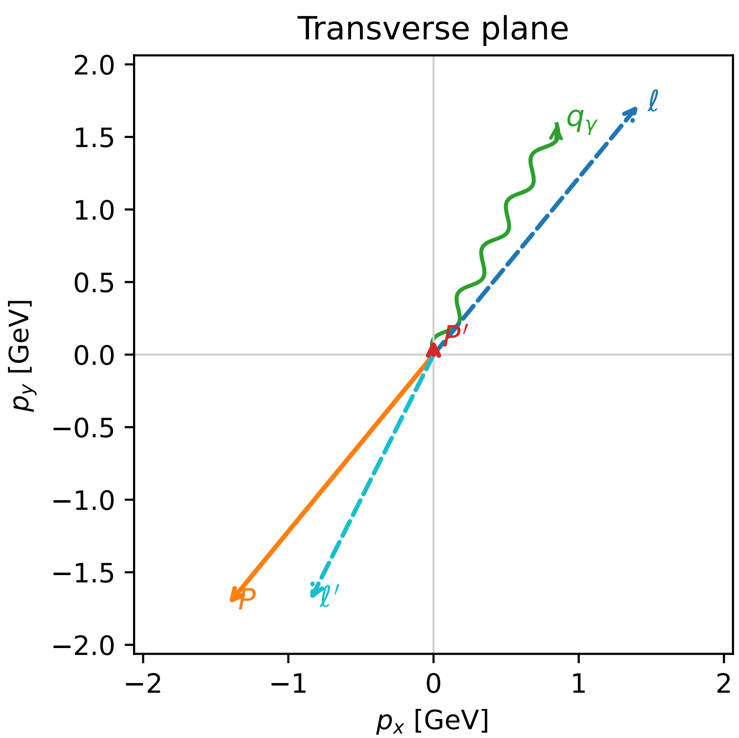
Tx proton characteristic max $C_{\mu p}$ configuration

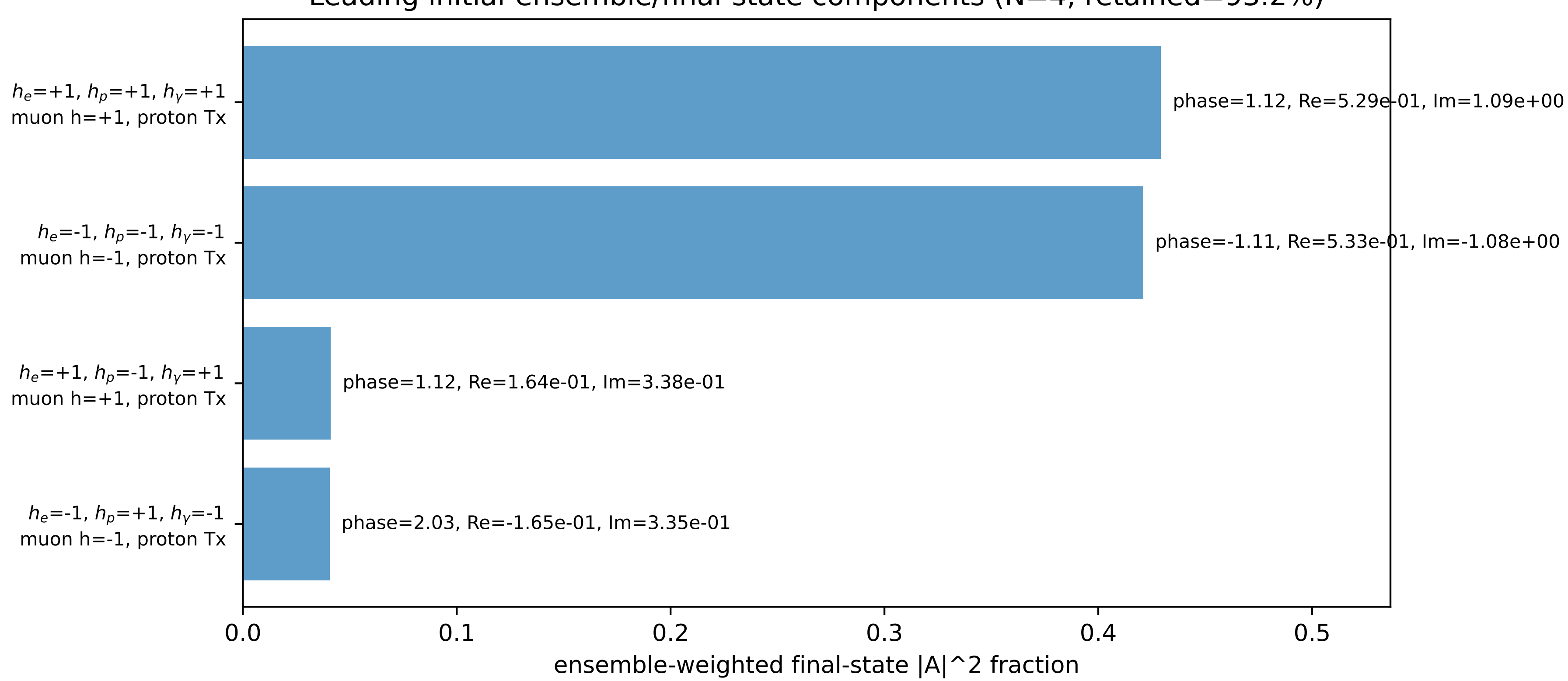


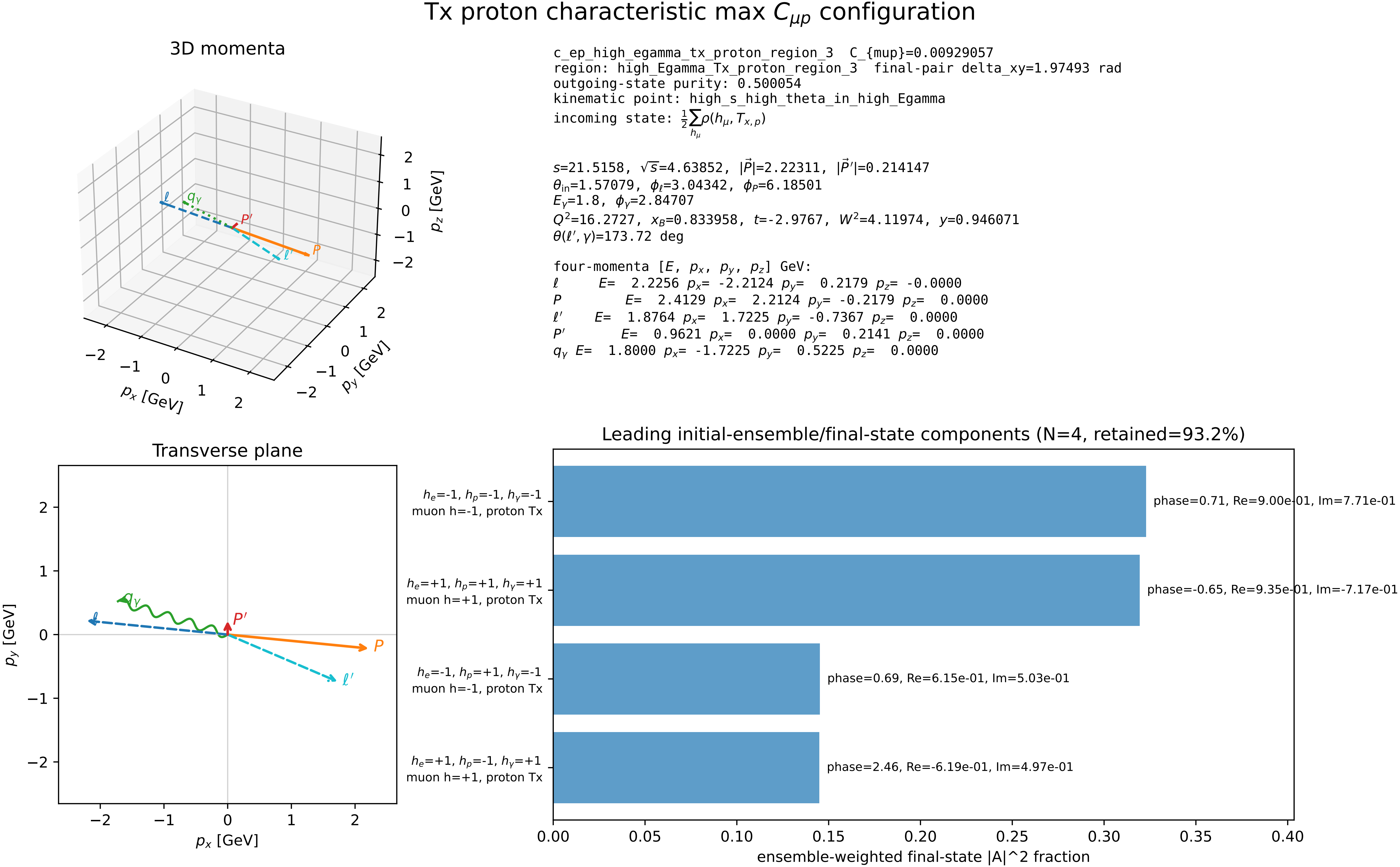
c_ep_high_egamma_tx_proton_region_2 C_{mup}=0.00942537
region: high_Egamma_Tx_proton_region_2 final-pair delta_xy=2.6765 rad
outgoing-state purity: 0.500038
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_\mu}\rho(h_\mu, T_{x,p})$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=0.103468
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0.883573, ϕ_P=4.02517
E_γ=1.8, ϕ_γ=1.07992
Q^2=16.6171, x_B=0.844112, t=-3.14998, W^2=3.94862, y=0.954472
$\theta(\ell', \gamma)$=178.523 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 1.4103 p_y= 1.7185 p_z= -0.0000
P E= 2.4129 p_x= -1.4103 p_y= -1.7185 p_z= 0.0000
ℓ' E= 1.8948 p_x= -0.8485 p_y= -1.6909 p_z= 0.0000
P' E= 0.9437 p_x= 0.0000 p_y= 0.1035 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.8485 p_y= 1.5875 p_z= 0.0000






Tx proton characteristic max $C_{\mu p}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
ℓ
q_γ
P'
ℓ'
P
c_ep_high_egamma_tx_proton_region_3 C_{mup}=0.00929057
region: high_Egamma_Tx_proton_region_3 final-pair delta_xy=1.97493 rad
outgoing-state purity: 0.500054
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_\mu}\rho(h_\mu, T_{x,p})$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=0.214147
θ_{in}=1.57079, ϕ_ℓ=3.04342, ϕ_P=6.18501
E_γ=1.8, ϕ_γ=2.84707
Q^2=16.2727, x_B=0.833958, t=-2.9767, W^2=4.11974, y=0.946071
$\theta(\ell', \gamma)$=173.72 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -2.2124 p_y= 0.2179 p_z= -0.0000
P E= 2.4129 p_x= 2.2124 p_y= -0.2179 p_z= 0.0000
ℓ' E= 1.8764 p_x= 1.7225 p_y= -0.7367 p_z= 0.0000
P' E= 0.9621 p_x= 0.0000 p_y= 0.2141 p_z= 0.0000
q_γ E= 1.8000 p_x= -1.7225 p_y= 0.5225 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=4, retained=93.2%)
h_e=-1, h_p=-1, h_γ=-1
muon h=-1, proton Tx
phase=0.71, Re=9.00e-01, Im=7.71e-01
h_e=+1, h_p=+1, h_γ=+1
muon h=+1, proton Tx
phase=-0.65, Re=9.35e-01, Im=-7.17e-01
h_e=-1, h_p=+1, h_γ=-1
muon h=-1, proton Tx
phase=0.69, Re=6.15e-01, Im=5.03e-01
h_e=+1, h_p=-1, h_γ=+1
muon h=+1, proton Tx
phase=2.46, Re=-6.19e-01, Im=4.97e-01
ensemble-weighted final-state |A|^2 fraction